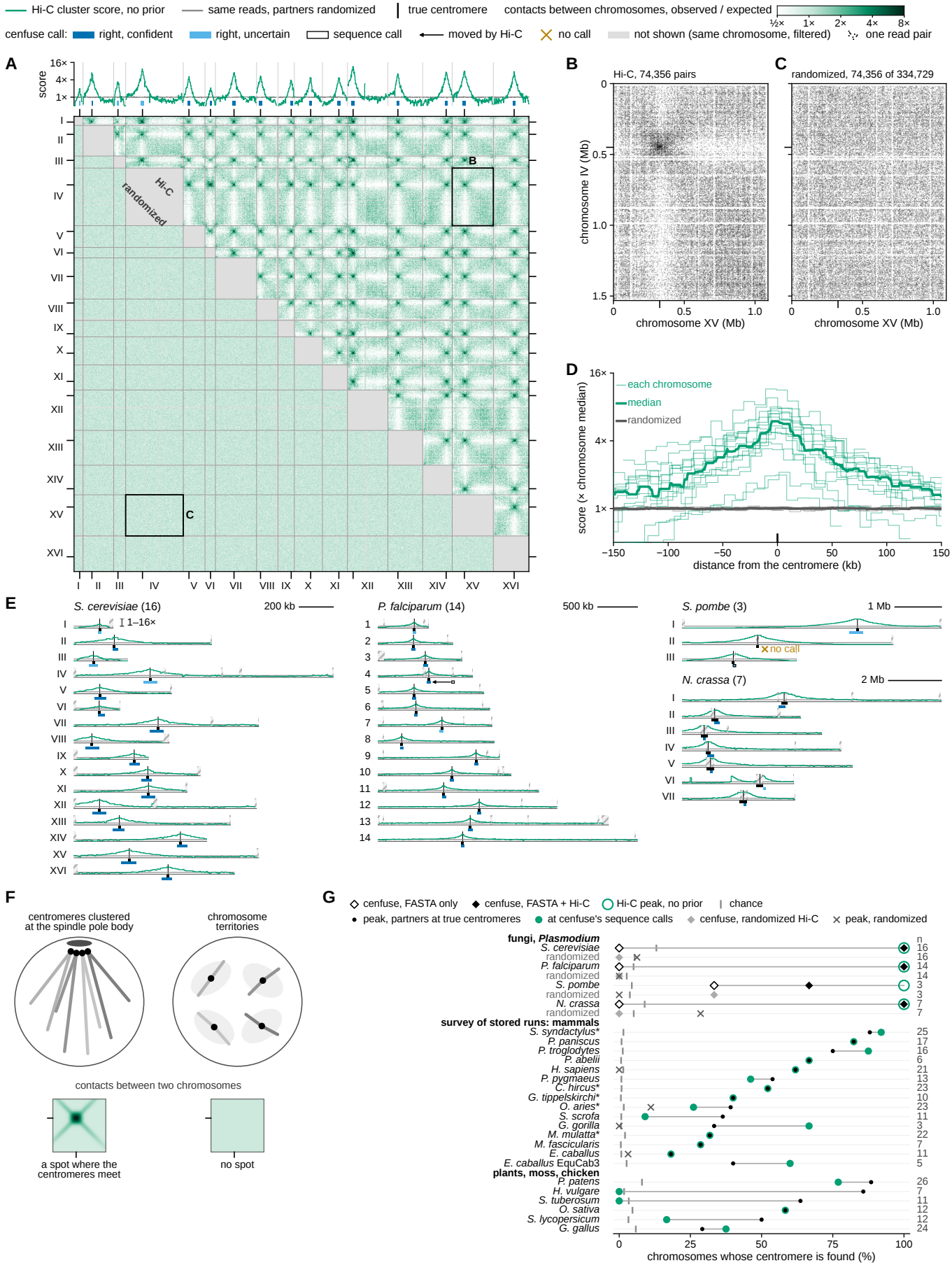

Hi-C cluster score, no prior
same reads, partners randomized
true centromere
contacts between chromosomes, observed / expected
½× 1× 2× 4× 8×
cenfuse call: right, confident
right, uncertain
sequence call
moved by Hi-C
no call
not shown (same chromosome, filtered)
one read pair
A
score
16×
4×
1×
Hi-C
randomized
B
C
I II III IV V VI VII VIII IX X XI XII XIII XIV XV XVI
B
Hi-C, 74,356 pairs
randomized, 74,356 of 334,729
chromosome IV (Mb)
chromosome XV (Mb)
0 0.5 1.0 1.5
D
each chromosome
median
randomized
score (× chromosome median)
distance from the centromere (kb)
−150 −100 −50 0 50 100 150
E
S. cerevisiae (16)
200 kb
1–16×
P. falciparum (14)
500 kb
S. pombe (3)
1 Mb
no call
N. crassa (7)
2 Mb
F
centromeres clustered at the spindle pole body
chromosome territories
contacts between two chromosomes
a spot where the centromeres meet
no spot
G
cenfuse, FASTA only
cenfuse, FASTA + Hi-C
Hi-C peak, no prior
chance
peak, partners at true centromeres
at cenfuse's sequence calls
cenfuse, randomized Hi-C
peak, randomized
fungi, Plasmodium
S. cerevisiae
randomized
P. falciparum
randomized
S. pombe
randomized
N. crassa
randomized
survey of stored runs: mammals
S. syndactylus*
P. paniscus
P. troglodytes
P. abelii
H. sapiens
P. pygmaeus
C. hircus*
G. tippelskirchi*
O. aries*
S. scrofa
G. gorilla
M. mulatta*
M. fascicularis
E. caballus
E. caballus EquCab3
plants, moss, chicken
P. patens
H. vulgare
S. tuberosum
O. sativa
S. lycopersicum
G. gallus
n
16 16 14 14 3 3 7 7 25 17 16 6 21 13 23 10 23 11 3 22 7 11 5 26 7 11 12 12 24
0 25 50 75 100
chromosomes whose centromere is found (%)

Where nuclei cluster their centromeres, Hi-C contacts alone find every centromere. Contacts between chromosomes mark all 40 centromeres of two yeasts, the malaria parasite and *Neurospora*, where sequence finds one; in mammals and plants they mark only some.

A, Contacts between the 16 chromosomes of budding yeast form a lattice of spots, one wherever two centromeres meet, as expected when all centromeres are held together at one pole of the nucleus. The same reads with their partners shuffled (lower half) show nothing; the score above peaks at every centromere.

B, Every read pair joining chromosomes IV and XV: they pile up where the two centromeres meet and fan out along the arms. **C**, The same block after shuffling: no focus.

D, All 16 yeast chromosomes aligned at their centromere: contacts with the other centromeres rise two- to twelvefold within a few tens of kilobases of it; shuffled data stay flat.

E, Every chromosome of four genomes: the Hi-C peak sits on the centromere on all 40, whether point, regional or AT-rich, and shuffled reads stay flat. From these contacts cenfuse calls 39 centromeres, 31 of them confidently, and moves one wrong sequence call onto its centromere; one chromosome stays uncalled.

F, Why it works: clustered centromeres touch each other, so every chromosome pair shares a contact spot; in chromosome territories they do not.

G, Sequence alone finds one of these 40 centromeres, cenfuse with Hi-C 39, shuffled Hi-C none beyond chance. In mammals, plants and chicken the contacts find only some centromeres, even when told where the partner centromeres are.

Terms. *Score* (green): for each bin, the mean contact enrichment (observed/expected) with windows of ±4 bins at the other chromosomes' current centromere estimates, iterated from the telomere-pole minimum until the windows stop moving; no sequence, annotation or prior position is used; shown as fold over the chromosome median (log). *Found* (right): the peak bin, or the call, lies within one Hi-C bin of the true centromere (250 kb for calls from the FASTA alone). *Confident*: cenfuse's level medium or high; *uncertain*: low, cenfuse's word for unsure. *Randomized*: the same read ends, partners re-paired at random. Calls: the cenfuse 0.10.1 release rerun of each genome's FASTA, alone and with its Hi-C pairs. Truth: *S. cerevisiae* CEN (SGD, 111–120 bp), *S. pombe* CENP-A/Cnp1 domain (15.3–15.6 kb), *P. falciparum* annotated AT-rich core (1.5–3.6 kb; chr10: CenH3 CUT&Tag), *N. crassa* CenH3 domain (174–288 kb).

A, *S. cerevisiae* BY4742 Hi-C, G1-arrested with alpha factor (14,861,297 read pairs, 19.6% between chromosomes), 10-kb bins, chromosomes I–XVI. Color: balanced contacts over the mean of their chromosome-pair block, so both halves share one scale although 92.5% of the randomized pairs fall between chromosomes. Gray: same-chromosome blocks and 36 bins removed by balancing. Black ticks: the 16 centromeres. All 120 chromosome pairs show a spot where their centromeres meet (3 × 3 bins: median 5.8×, interquartile 5.25–6.32×, range 2.42–8.27×; randomized 1.01×) and crosses along the arms, the contact pattern of centromeres held at the spindle pole body with arms trailing side by side (Rabl configuration). Track: the score of the real (green) and randomized (gray) library; boxes: cenfuse's 16 calls, all on the centromere (13 confident, dark blue; 3 uncertain, light blue).

B, Every read pair between chromosome IV (1.53 Mb) and XV (1.09 Mb), n = 74,356 (mapping quality ≥ 30 at both ends; equal to the map's block), each a dot at its two positions; black ticks: CEN4, CEN15. The two longest chromosomes were chosen for their long arms; their spot is at the 81st percentile of the 120 pairs (6.52×), typical rather than the strongest. White stripes are stretches without uniquely mapped reads, shared with C because randomization keeps every read end in place.

C, The same block in the randomized library: 74,356 of its 334,729 pairs drawn at random so both panels hold equally many dots, drawn in B's orientation (its box in A is the mirror block below the diagonal). No focus: the spot needs partners that touched in the nucleus.

D, The score of each yeast chromosome (thin) and the median (bold) aligned at the centromere midpoint, against the randomized library (gray). Peaks reach 2.0–11.6× the chromosome median (randomized at most 1.08×); the half-maximum core spans a median 65 kb. Scores sink below the median toward the chromosome ends, far from the cluster.

E, Every chromosome of the four genomes, to scale within each genome: score (green, log, 0.5–16×) and the randomized library's score (gray), true centromere (black bar; thin line at its middle), cenfuse's call (dark blue: confident; light blue: uncertain; outlined: a sequence call; at least 0.5 mm wide), hatched: bins removed by balancing. Bins 10 kb (*S. cerevisiae*, *P. falciparum*: G1 cells, ring-stage parasites) or 50 kb (*S. pombe*, *N. crassa*: asynchronous cells, germinated conidia). The peak is found on 16/16, 14/14, 3/3 and 7/7 chromosomes, at most 6.3, 2.0, 42.7 and 0 kb from the centromere; randomized scores never exceed 1.32× their median. With too few sequence calls to test Hi-C against, cenfuse tests it on the Rabl axis: calls are confident where the axis test passed and holds the peak (trans arm alignment: yeast 13/16, *P. falciparum* 12/14; arm fold: *N. crassa* 6/7), uncertain elsewhere and in *S. pombe*, where no test passed. *P. falciparum* chr4: its sequence call (hollow), two tandem copies of an 11.9-kb unit at 939–963 kb, is moved 269 kb onto the centromere (620–670 kb). *N. crassa* LG VI: the uncertain call (3.05–3.15 Mb) sits at the edge of the CenH3 domain (2.81–3.06 Mb), beside an inversion in the RefSeq assembly (1.85–2.79 Mb). *S. pombe* chr II stays uncalled: its peak, 18.1 kb from CENP-A, is only of medium strength, and Hi-C alone calls only at strong peaks; chr III keeps its sequence call, a tandem repeat of a 6.8-kb unit 2.7 kb from CENP-A.

F, Schematic, not data. Left: the Rabl configuration, centromeres (black) held at the spindle pole body or microtubule-organizing center through interphase; right: chromosome territories. Lineages without condensin II tend to keep the Rabl configuration (Hoencamp et al. 2021).

G, Each row a genome; n: chromosomes scored. Top, the four genomes of E: cenfuse on the FASTA alone (0/16, 0/14, 1/3 and 0/7), with Hi-C (16/16, 14/14, 2/3 and 7/7; 31 of the 39 right calls confident, none wrong; $P = 2.0 \times 10^{-15}$, 2.2×10^{-19}, 0.0043 and 3.8×10^{-8}) and the prior-free peak (40/40). Chance (tick): right calls expected by luck from the widths of calls and centromeres (2.09, 0.71, 0.13 and 0.63 chromosomes; 0.87, 0.37, 0.11 and 0.34 for a one-bin peak). Randomized rows: no test passes and cenfuse keeps the FASTA-only calls; the score's maximum lands on 3 of 40 centromeres (1.7 expected, P = 0.24; *N. crassa* alone 2/7, P = 0.046, none of them a peak the engine reports). All four genomes show the telomere-to-centromere polarity of a Rabl nucleus (contacts with the other chromosomes' ends fall with distance from the own end on 16/16, 14/14, 3/3 and 7/7 chromosomes; yeast sign test $P = 1.5 \times 10^{-5}$); their randomized libraries do not (8/16, 5/14, 2/3 and 2/7). Below: a survey of stored runs of the earlier benchmark (not rerun for this release): 14 T2T mammals, EquCab3 horse, four plants, moss and chicken (250-kb bins; human 500 kb, barley 2 Mb), chromosomes whose centromere lies next to mappable bins (found: within one mappable bin, up to four unmappable bins free). Black: the peak with partner windows at the true centromeres (an oracle, the best these contacts allow); green: at cenfuse's sequence calls. The oracle finds a median 46% in T2T mammals (18–88%, apes highest); sequence anchors fail in potato (0/11), barley (0/7) and tomato (2/12), where the oracle finds 7/11, 6/7 and 6/12: there the anchors fail, not the contacts. Randomized (×): 4 of 91 chromosomes (3.4 expected). Asterisk: truth is the satellite-array locus, not CENP-A. The holocentric *C. elegans* gets no confident call (6 low calls, as from the FASTA alone or with randomized Hi-C) and lacks the telomere polarity (1/6, P = 0.98).

Yeast is the showcase: its point centromeres (111–120 bp) carry no sequence a caller can find, and with 16 chromosomes it has the most centromere pairs (120) of the four genomes; the chromosome pair in B and C was chosen by length, not by signal.

Analysis: asc-oulab:/data2/chris/centromere_benchmark/v2/v3/v4/v5/v6/v7/v8/v9/v10/v10.1/v10.2 · Script: /data2/chris/centromere_benchmark/v2/v3/v4/v5/v6/v7/v8/v9/v10/v10.1/v10.2/figures/benchmark_hic/benchmark_hic.py · 2026-11-04